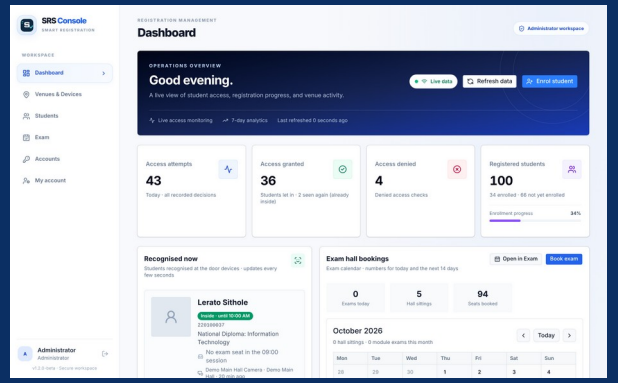
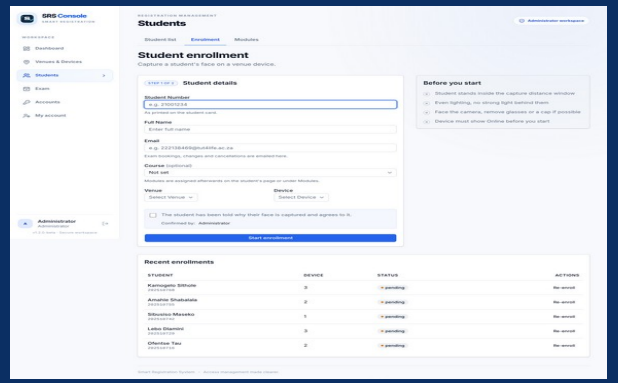
Tshwane University of Technology

*We empower people*

# Smart Registration System

Face-recognition access control for university exam venues




TC Mance · 222138469 · Project Design 376 (PD2316D)
Lecturers: Mr M. Muwanguzi Ntume and Prof C. Du · 5 November 2026

# Checking identity by eye is slow and easy to fool

### Slow at the door

Every candidate is compared with a register and a student card

### Interrupts writing

Checks are repeated during the exam, disturbing students

### Impersonation slips through

A convincing look-alike passes a quick visual check

### No reliable record

Paper ticks give no time-stamped audit trail

## The aim

Verify every candidate automatically at the venue door

Show each student's seat and session at the door

Keep the invigilator in charge of every refusal

# Four objectives from the proposal – all met

## Fast, distance-gated recognition

MET

Goal: Face verified within 2 s at 90%+ accuracy

**≈1 s hold at 25–50 cm, then the face is matched**

## Low FAR and FRR

MET

Goal: Keep both error rates acceptable

**Strict 0.40 match threshold · one face per student**

## Log every attempt, live dashboard

MET

Goal: Time, student and decision for each entry

**Every decision stored; live console for all venues**

## Manual override

MET

Goal: Backup when recognition fails

**Override button records who and why**

# Face recognition exists – exam-aware access does not

## Commercial products

Facephi · SmartVisitor · Navon · Openitem

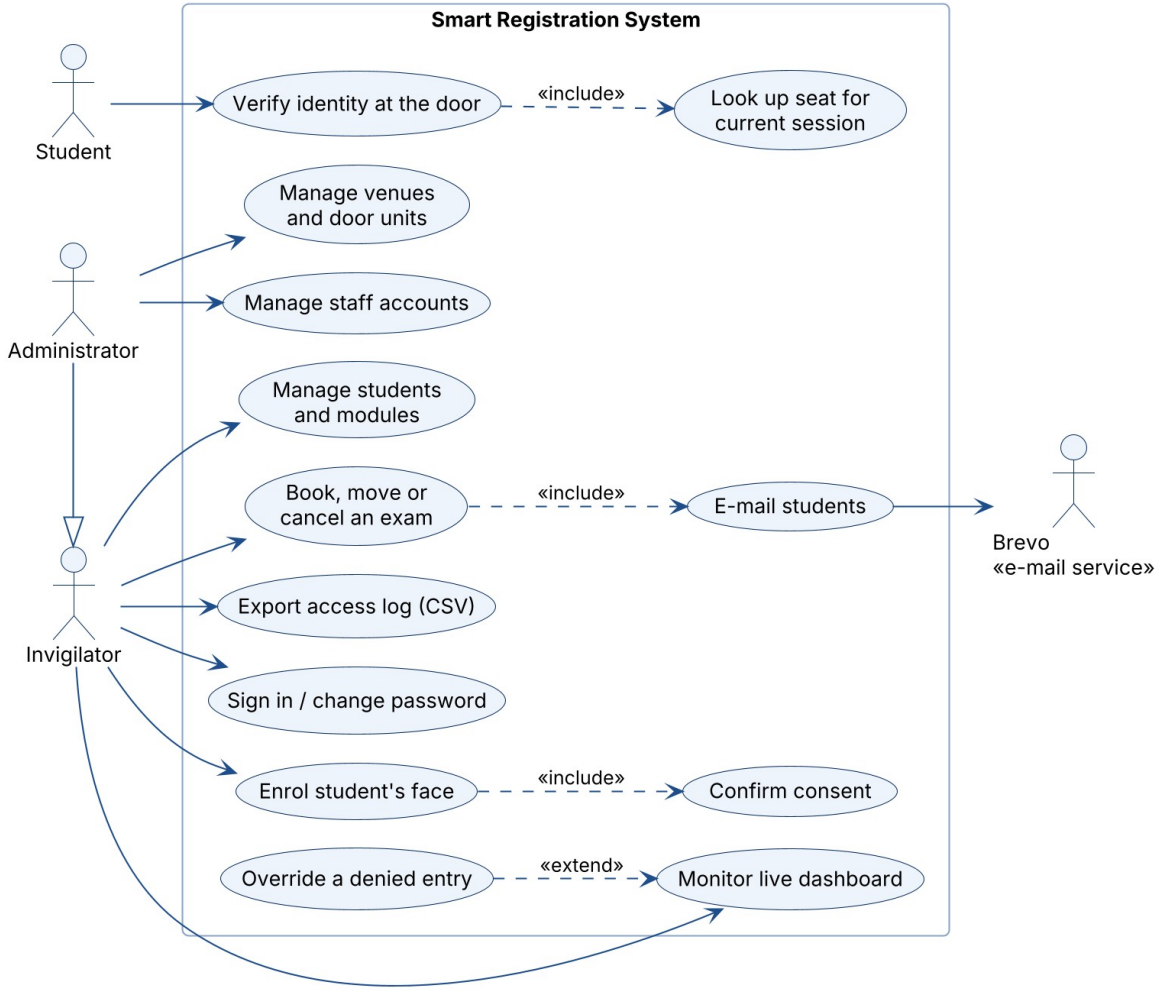
- Offices, estates, events and visitors
- No idea of exam timetables or seats
- Face templates usually kept in the cloud
- Subscription – about R480 per month

## Smart Registration System

Built for TUT exam venues

- **Shows each student's seat; flags "No seat" for the invigilator**
- **Once per session; automatic sign-out at 12:00 / 16:00**
- **Face templates stay on the door PC (POPIA)**
- **About R1 225 of hardware, free hosting tiers**

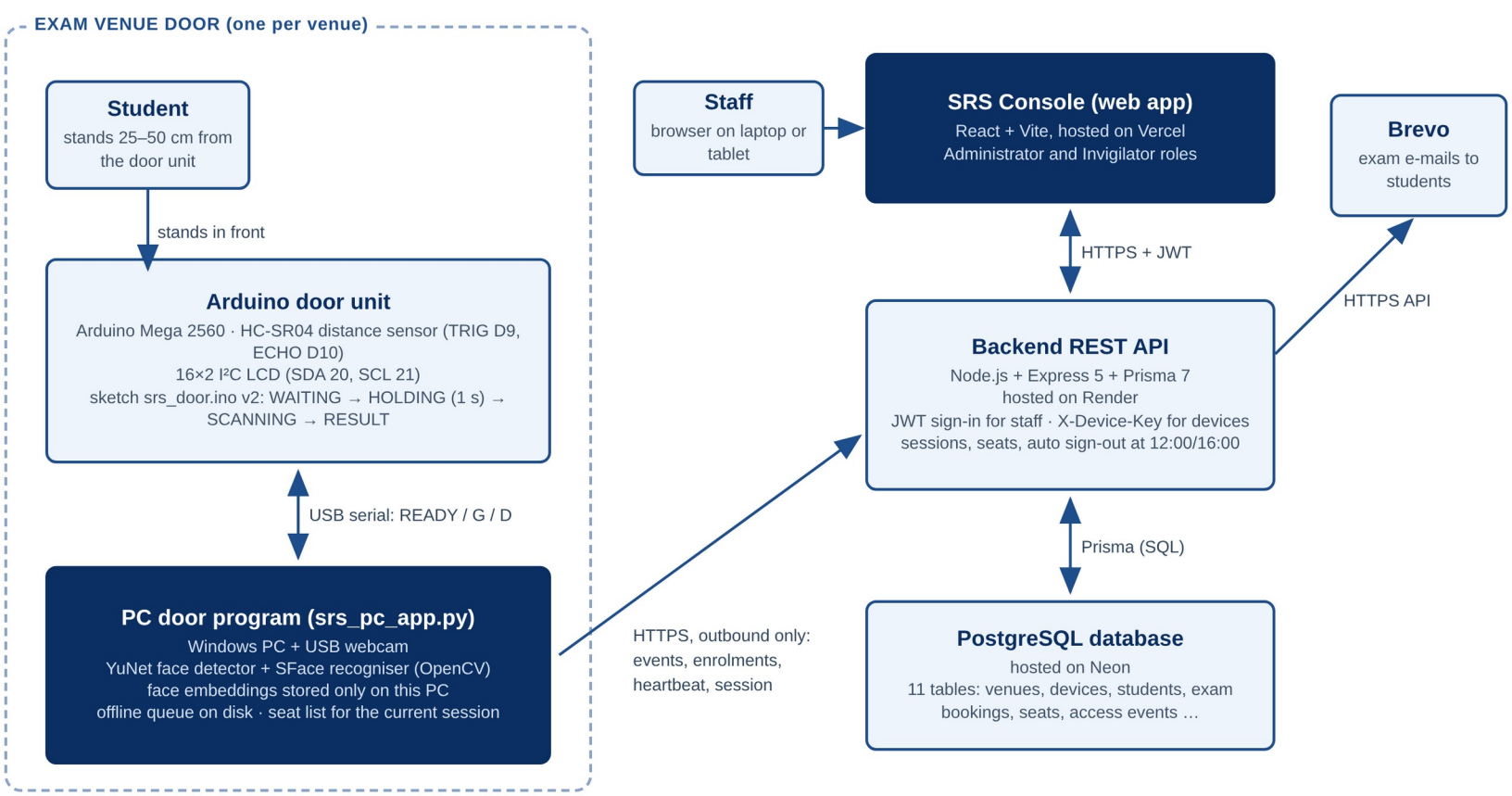
# Who uses the system: UML use case diagram




## What it shows

- Invigilators enrol faces, manage students, book exams, watch the dashboard and export the log
- Administrators can also manage venues, door units and staff accounts
- Students only verify their identity at the door, which looks up their seat
- Enrolment includes consent; booking an exam e-mails the students (Brevo)
- Invigilators can override a denied entry from the dashboard

# Four parts, connected only through the backend




## Door unit

PC + webcam recognises; Arduino measures distance, shows the result

## Backend

Node.js + Express on Render; decides sessions, seats, sign-out

## Security

Staff sign in (JWT); doors use a secret device key; outbound HTTPS only

# The hardware change: ESP32-CAM to PC + Arduino

## Planned

- ESP32-CAM with OV2640 camera
- ESP-WHO recognition on the board
- Battery pack + buck converter
- Status LED for feedback

## Final

- **USB webcam on a Windows PC**
- **OpenCV YuNet + SFace recognition**
- **USB power – no batteries**
- **Arduino Mega: distance sensor and LCD**

*Why: hardware problems with the ESP32-CAM could not be solved in time. The new design removed its memory, power and pin limits – and the new door unit uses the same backend API.*

# The door unit waits for the student to hold still




**25–50 cm**

distance window (HC-SR04)

**1 s**

hold before the picture

**G / D / A**

answer shown on the LCD

*d = 343 m/s × t ÷ 2 → 25–50 cm = 1.45–2.9 ms echo*

# From face to decision: match first, then session rules

### 1. Hold

Student holds still 1 s at 25–50 cm

### 2. Detect

YuNet finds the largest face

### 3. Match

SFace: 128 numbers, compared with ALL faces, ≥ 0.40

### 4. Session

Already inside? Session over? Seat looked up

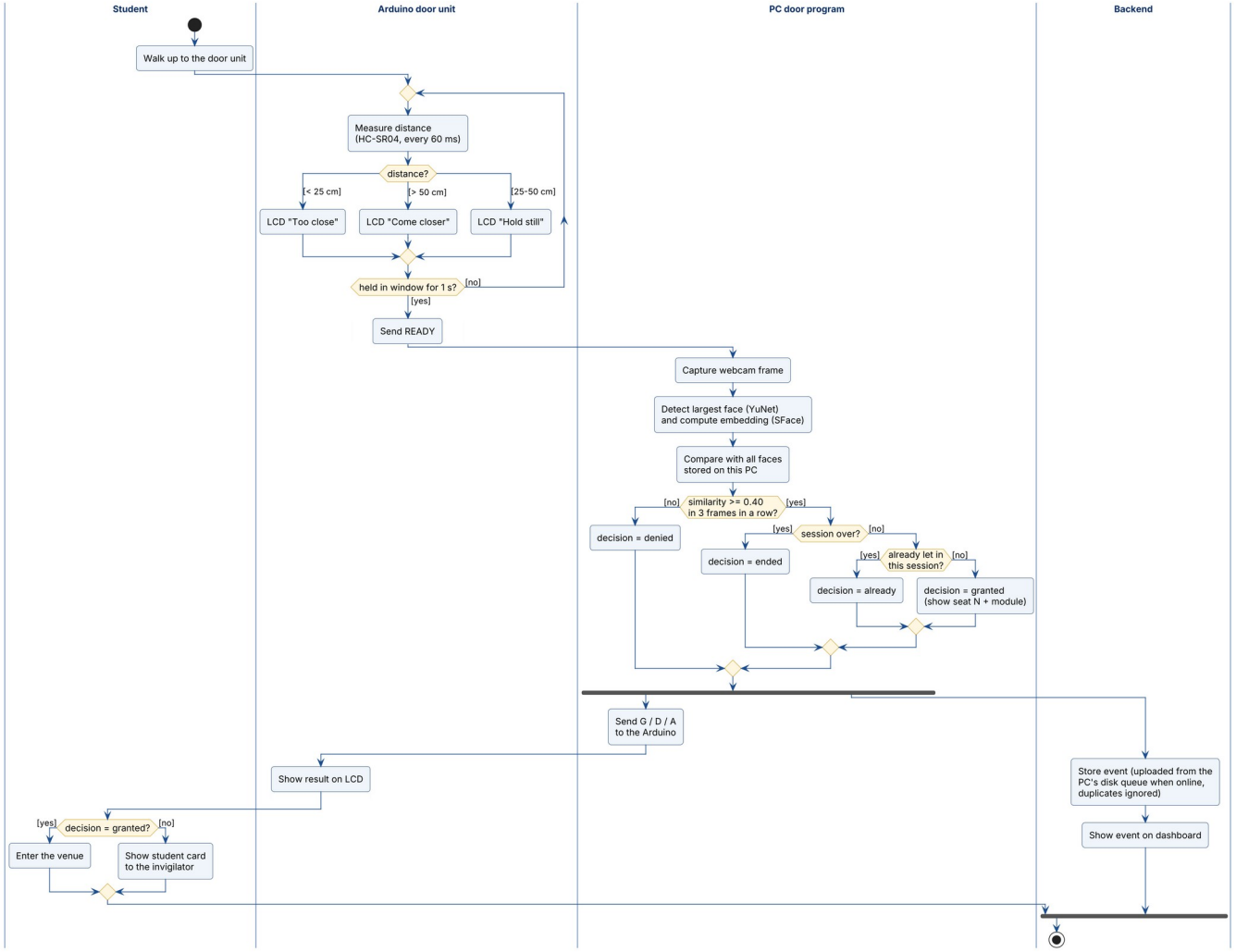
### 5. Decide

Let in with seat N (or "no seat" flag); unknown face DENIED

Offline? Decisions continue; events are queued on disk and uploaded once, never twice

Face templates stay on the door PC; the backend only gets the result and one enrolment photo

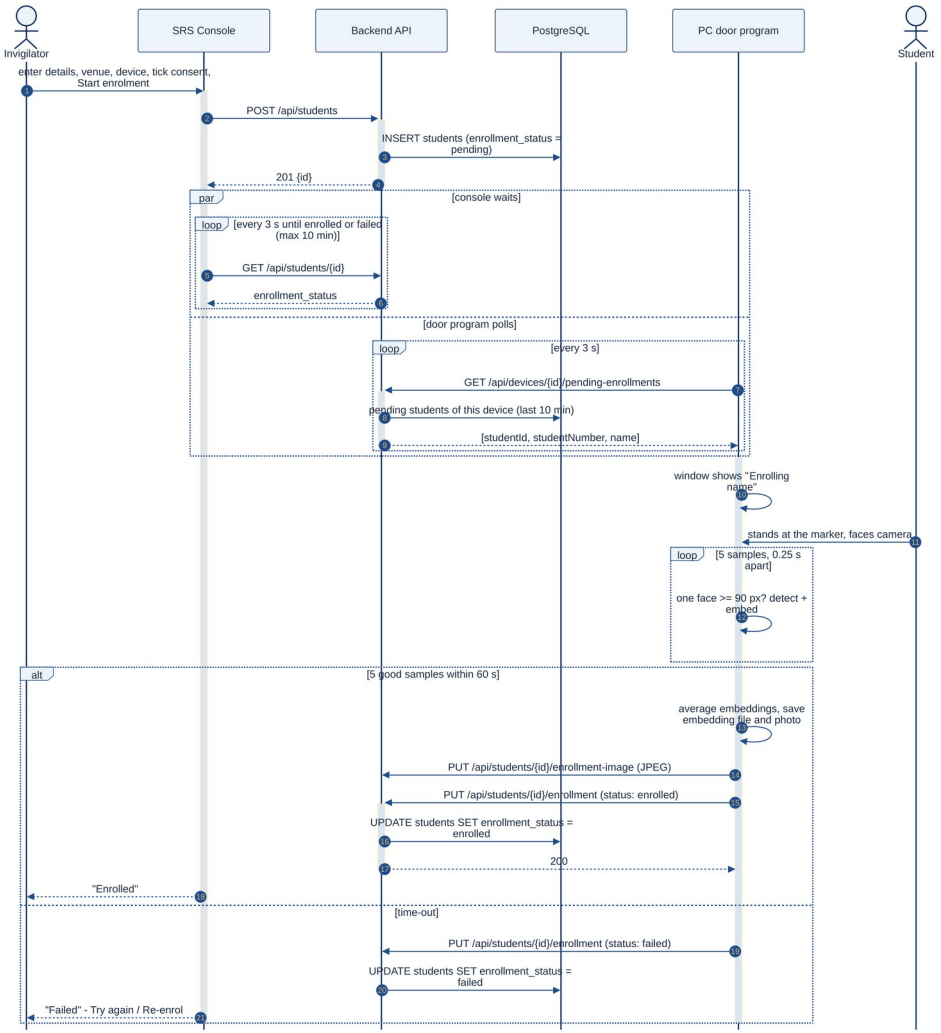
# Access verification: UML activity diagram




## What it shows

- The Arduino measures the distance every 60 ms and shows "Too close", "Come closer" or "Hold still"
- After a 1 s hold at 25–50 cm it sends READY to the PC
- The PC detects the face, compares it with every stored face and decides: denied, ended, already or granted
- The result goes to the LCD while the backend stores the event and shows it on the dashboard
- A student who is not let in shows their card to the invigilator

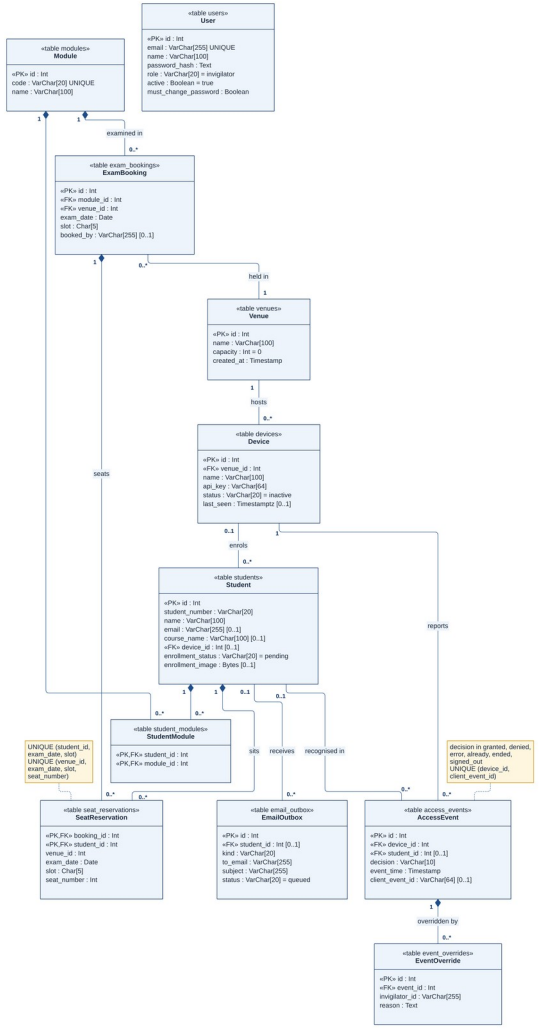
# Enrolment: UML sequence diagram




## What it shows

- The invigilator enters the student, ticks consent and starts enrolment; the student is saved as pending
- The console and the door program both check the backend every 3 s
- The door program takes 5 face samples, 0.25 s apart
- Success: the embedding stays on the PC, the photo is uploaded and the student becomes enrolled
- No 5 good samples in 60 s: the status becomes failed and the invigilator can try again

# Database: UML class diagram (11 tables)




## What it shows

- 11 tables; each box lists a table's columns and data types
- A venue hosts door units; a door unit enrols students and reports access events
- Every entry decision is an access event, which an invigilator can override
- Modules are examined in venues through exam bookings; each booking reserves a seat per student
- Students register for modules and receive exam e-mails; staff accounts stand alone

# SRS Console: one app for invigilators and admins

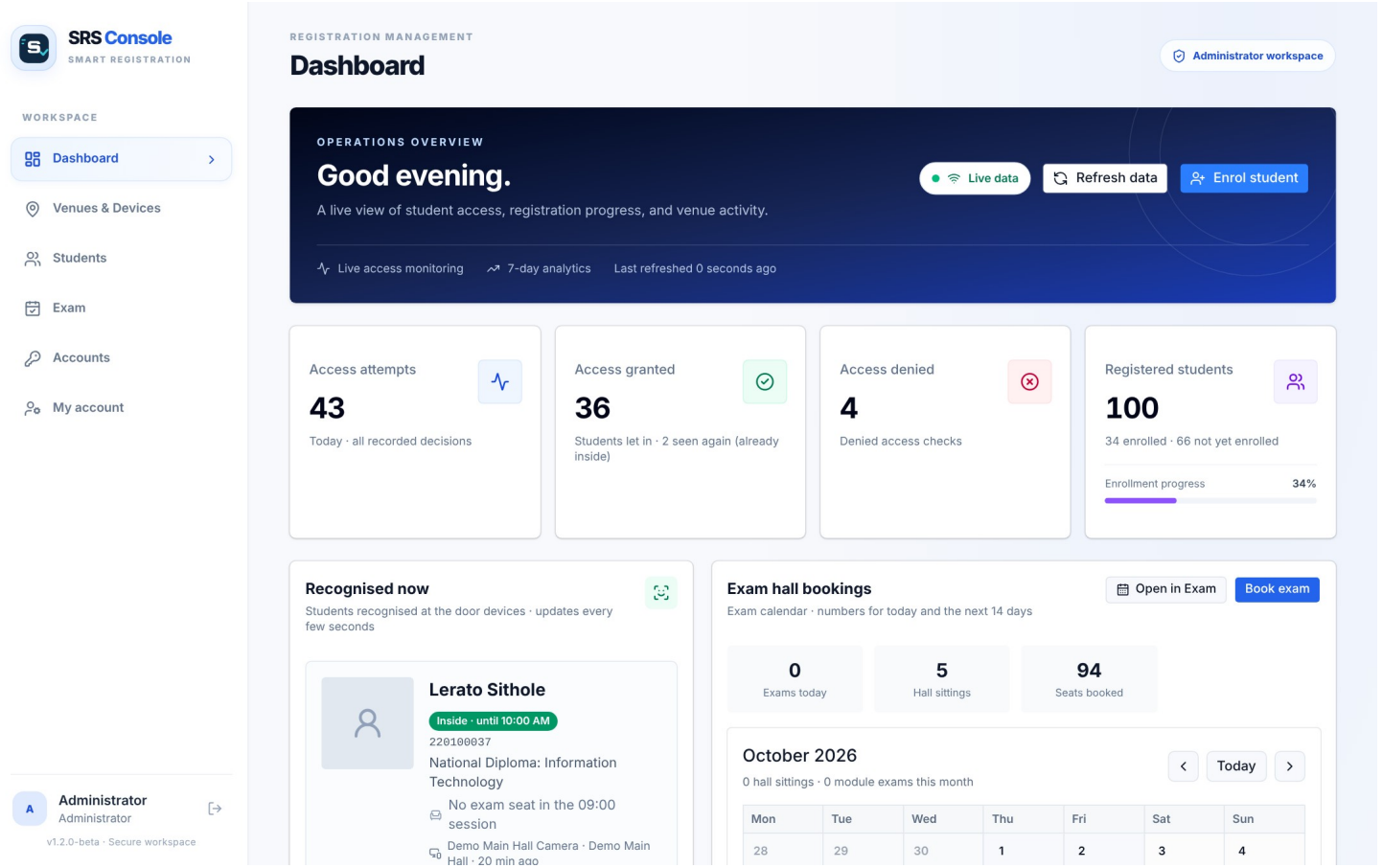
### Live dashboard
entries, refusals, recognised now

### Enrolment
with consent tick box

### Exam bookings
seats, capacity, calendar

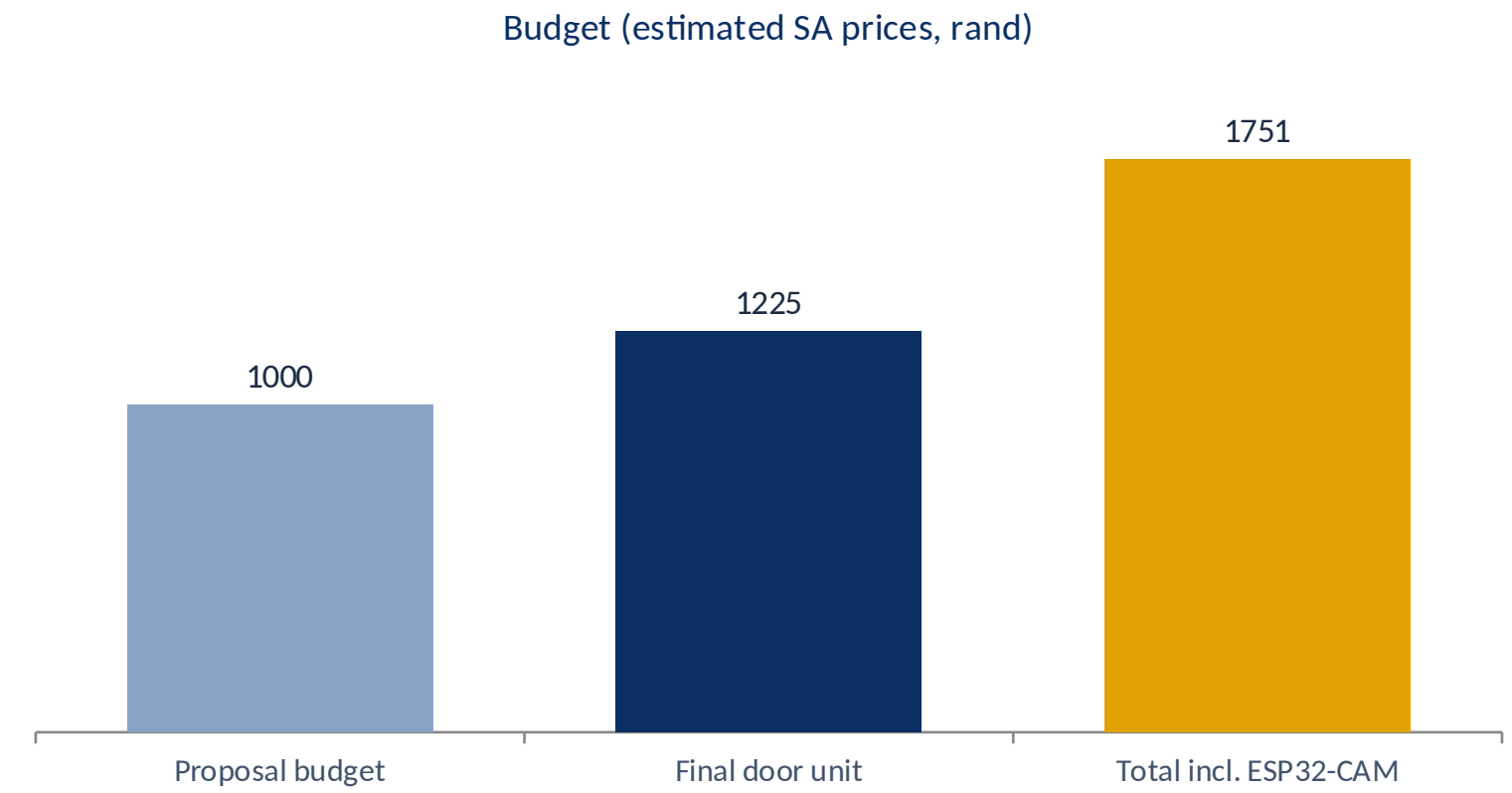
### E-mails
booked, moved, cancelled

### Two roles
admin and invigilator




Demo data shown

# Costed in rand




*Over budget because two sets of hardware were bought after the ESP32-CAM was dropped.*

# Conclusion and next steps

## What was achieved

- All four proposal objectives met
- Recognition tied to exam bookings and seats
- Biometric templates stay on the door unit
- Grew to many venues, e-mails and staff roles

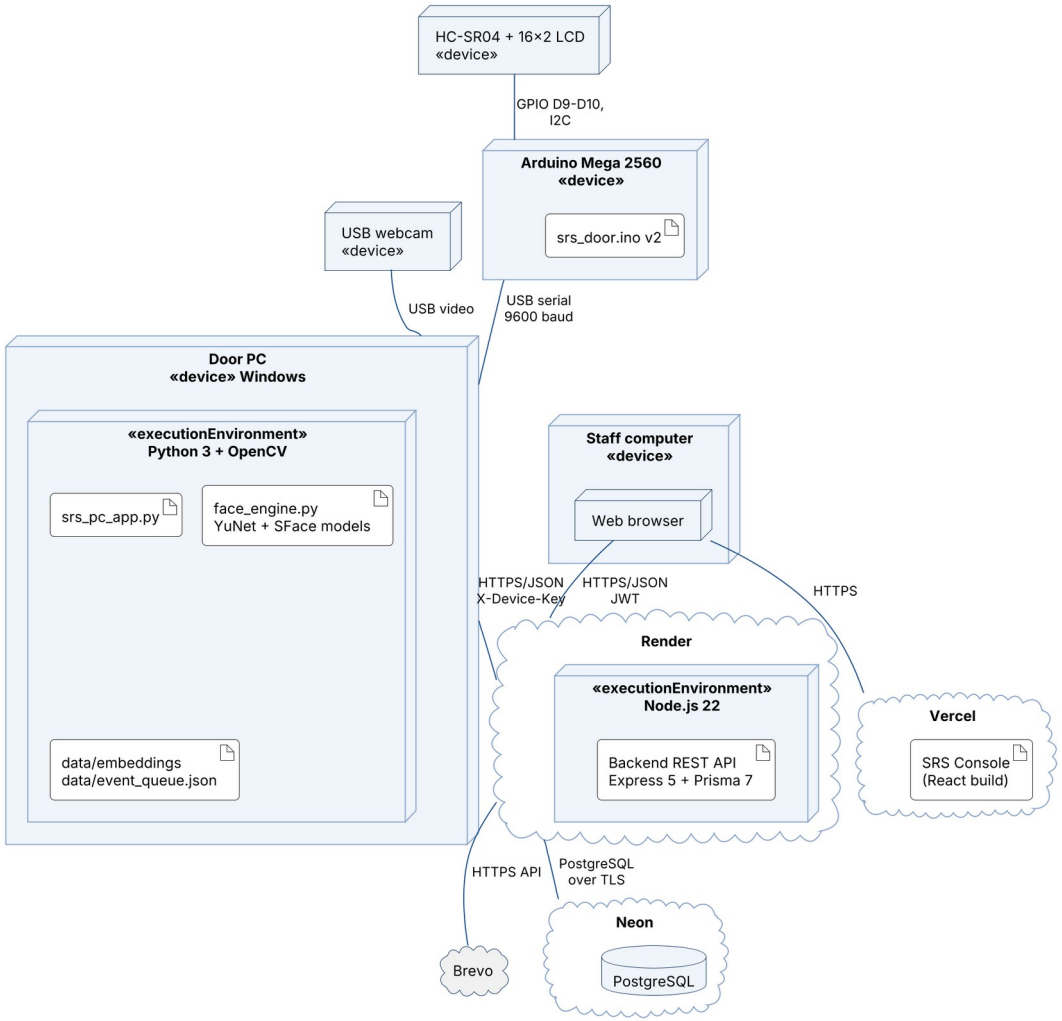
## Next steps

- On-site FAR/FRR test with 30+ volunteers
- Liveness check against printed photos
- Move the door program to a Raspberry Pi 5
- Import students from the university system

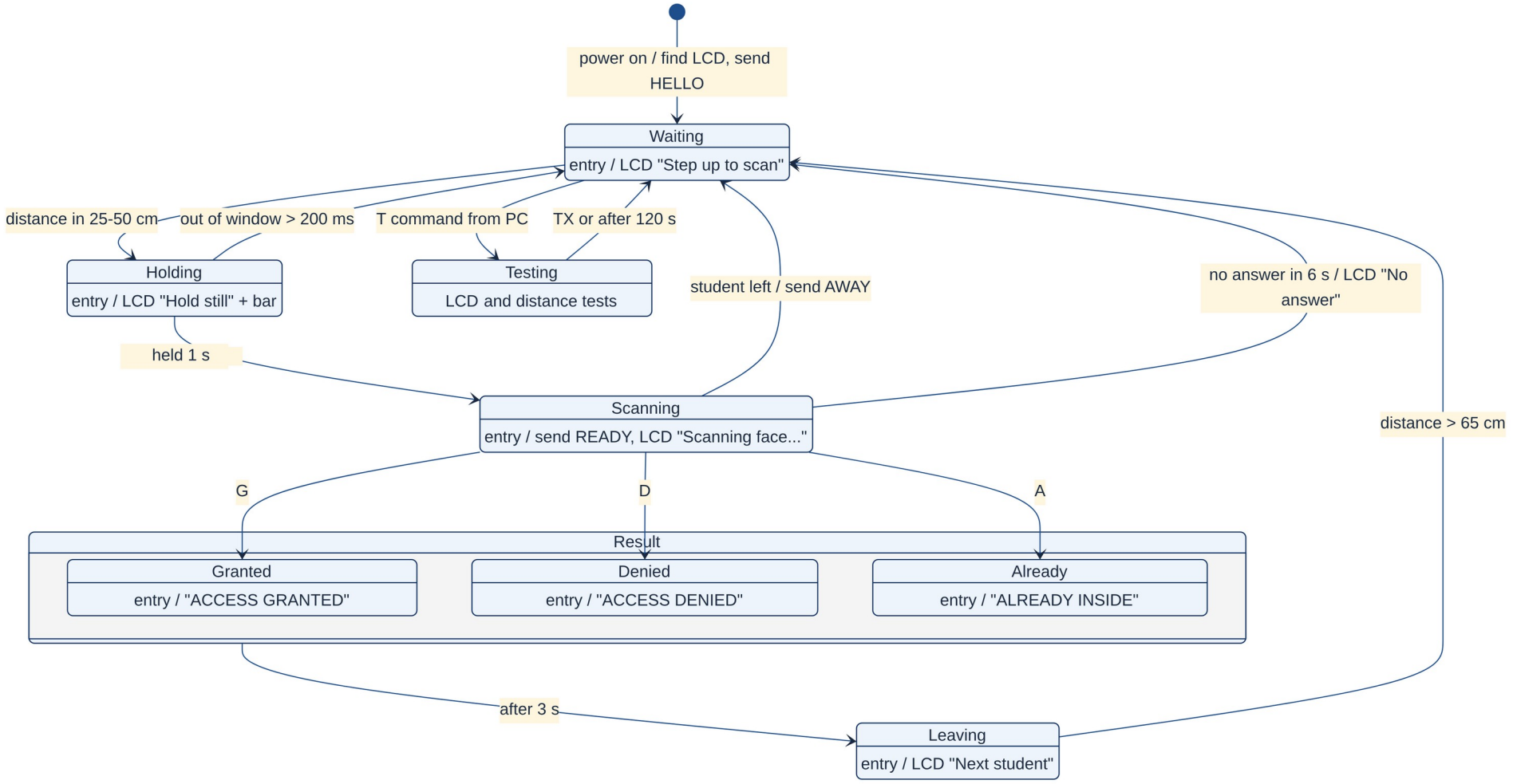


**Thank you – live demonstration and questions**

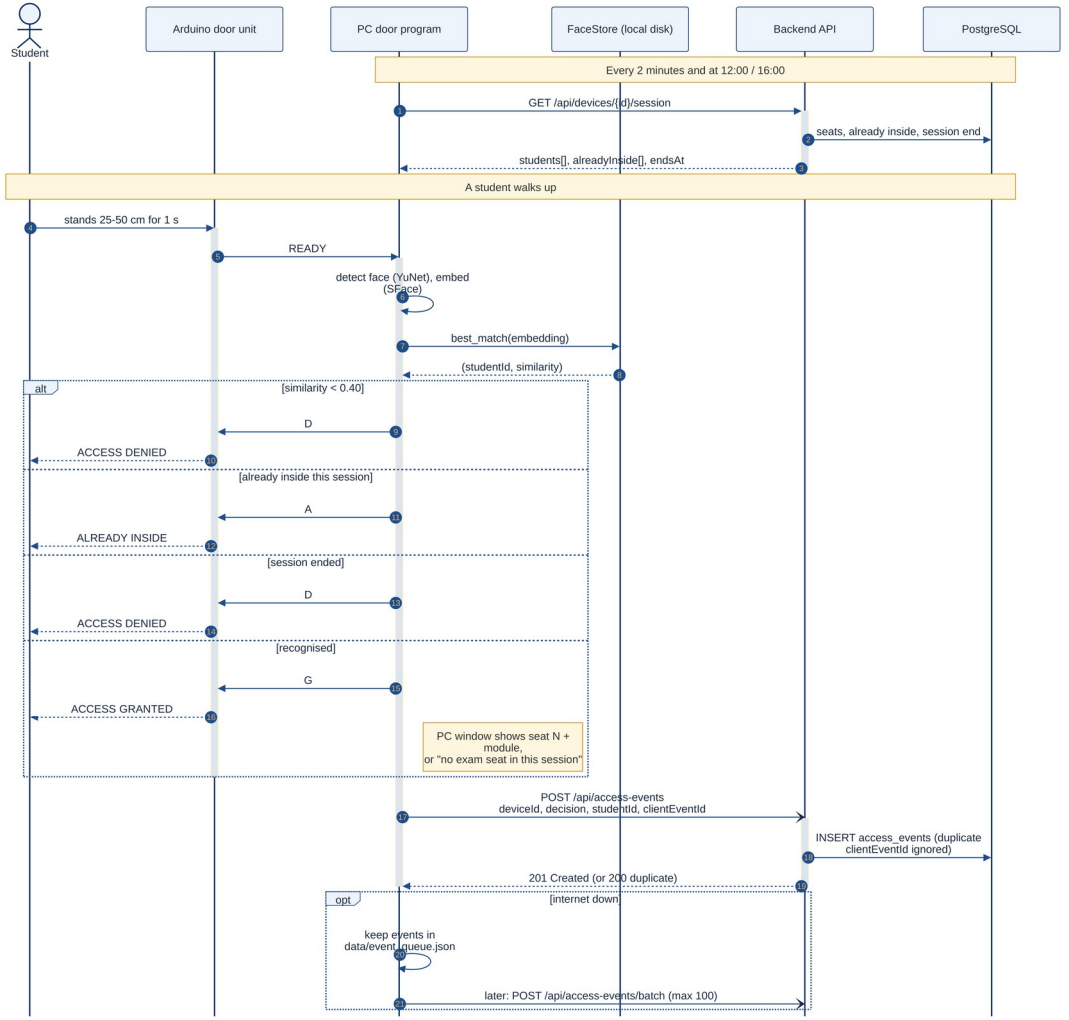
# Backup: UML deployment diagram



# Backup: UML state machine of the door unit



# Backup: UML sequence – one access verification



# Backup: UML sequence – booking, e-mails, sign-out

